

ONLINE MASTERS IN **DATA SCIENCE**

DSC 257R - UNSUPERVISED LEARNING

THE MULTIVARIATE GAUSSIAN

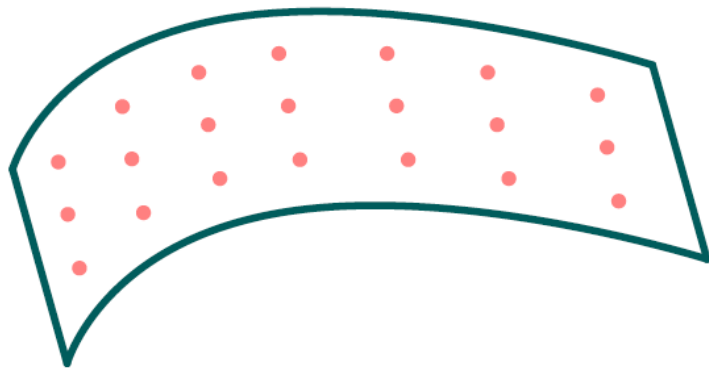
SANJOY DASGUPTA, PROFESSOR

UC San Diego

COMPUTER SCIENCE & ENGINEERING
HALICIOĞLU DATA SCIENCE INSTITUTE

Example: Modeling a Network of Sensors

Increasingly, large structures (bridges, skyscrapers) contain arrays of embedded sensors to warn of excessive strain or small cracks.

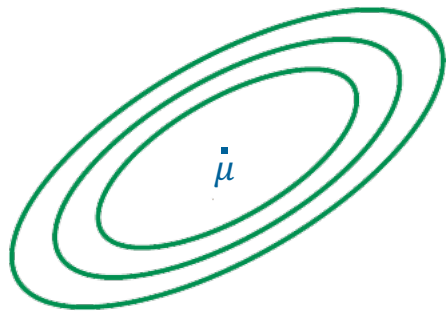


Say there are d sensors, each with a real-valued reading

- 1 Model distribution of sensor readings under normal conditions.
- 2 Accommodate a moderate number of sensor failures.

How to exploit correlations between sensors?

The Multivariate Gaussian



$N(\mu, \Sigma)$: Gaussian in $\mathbb{R}^d$

- mean: $\mu \in \mathbb{R}^d$
- covariance: $d \times d$ matrix Σ

Generates points $X = (X_1, X_2, \dots, X_d)$.

- μ is the vector of coordinate-wise means:

$$\mu_1 = \mathbb{E}X_1, \mu_2 = \mathbb{E}X_2, \dots, \mu_d = \mathbb{E}X_d.$$

- Σ is a matrix containing all pairwise covariances:

$$\begin{aligned}\Sigma_{ij} &= \Sigma_{ji} = \text{cov}(X_i, X_j) \text{ if } i \neq j \\ \Sigma_{ii} &= \text{var}(X_i)\end{aligned}$$

$$\text{Density } p(x) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu)\right)$$

Special Case: Spherical Gaussian

The X_i are independent and all have the same variance σ^2 .

1. What is the covariance matrix Σ , and what is its inverse Σ^{-1} ?

2. Simplify the density

$$p(\mathbf{x}) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2} (\mathbf{x} - \mu)^T \Sigma^{-1} (\mathbf{x} - \mu)\right)$$

Spherical Gaussian: Summary

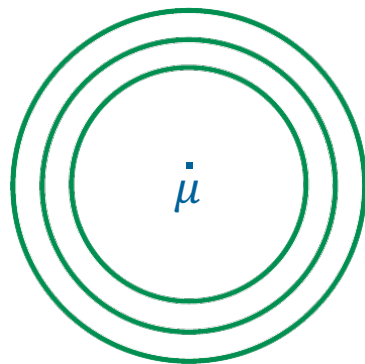
The X_i are independent and all have the same variance σ^2 .

$$\Sigma = \sigma^2 I_d = \text{diag}(\sigma^2, \sigma^2, \dots, \sigma^2) \quad (\text{diagonal elements } \sigma^2, \text{ rest zero})$$

Each X_i is an independent univariate Gaussian $N(\mu_i, \sigma^2)$:

$$\Pr(x) = \frac{1}{(2\pi)^{d/2} \sigma^d} \exp\left(-\frac{\|x - \mu\|^2}{2\sigma^2}\right) = \prod_{i=1}^d \left(\frac{1}{\sigma\sqrt{2\pi}} e^{-(x_i - \mu_i)^2 / 2\sigma^2}\right)$$

Density at a point depends only on its distance from μ :



Special Case: Diagonal Gaussian

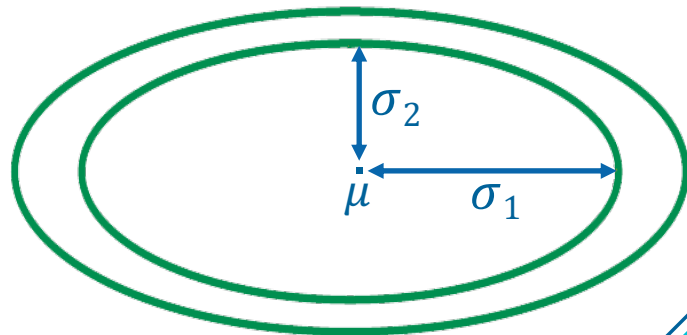
The X_i are independent, with variances σ_i^2 . Thus

$$\Sigma = \text{diag}(\sigma_1^2, \dots, \sigma_d^2) \text{ (off-diagonal elements zero)}$$

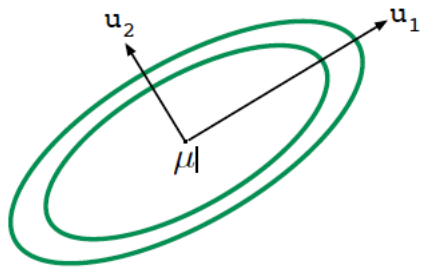
Each X_i is an independent univariate Gaussian $N(\mu_i, \sigma_i^2)$:

$$p(x) = \frac{1}{(2\pi)^{d/2} \sigma_1 \cdots \sigma_d} \exp\left(-\sum_{i=1}^d \frac{(x_i - \mu_i)^2}{2\sigma_i^2}\right)$$

Contours of equal density are **axis-aligned ellipsoids** centered at μ :



The General Gaussian $N(\mu, \Sigma)$ in $\mathbb{R}^d$



$$p(x) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp \left(-\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu) \right)$$

- **Eigenvectors** of Σ are $u_1, \dots, u_d$
- Corresponding **eigenvalues** $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_d$

Rotated version of a diagonal Gaussian $N(\mu, \Lambda)$: